Step 1 (Laspeyres)

$$Y_1^{(L)} = p_1^F q_1^F + p_1^C q_1^C = \$200$$

$$Y_2^{(L)} = p_1^F q_2^F + p_1^C q_2^C = (\$6 \times 25) + (\$8 \times 20)$$

$$= \$310$$

$\Rightarrow$ Growth in $Y^{(L)}$ between Year 1 and 2:

$$g^L = \frac{\$310}{\$200} = 1.55$$

Step 2 (Paasche)

$$Y_1^P = P_2^F q_1^F + P_2^C q_1^C$$

$$= (\$10 \times 20) + (\$7 \times 10)$$

$$= \$270$$

$$Y_2^P = P_2^F q_2^F + P_2^C q_2^C$$

$$= \$390$$

Growth in Y using this method:

$$g^P = \frac{390}{270} = \underline{1.44}$$

↑

Gross growth (growth factor)

Step 3 Take "geometric" average of g^L and g^P:

$$g^F = (g^L)^{\frac{1}{2}}(g^P)^{\frac{1}{2}}$$

$$= (1.58)^{\frac{1}{2}}(1.44)^{\frac{1}{2}}$$

$$= 1.49$$

STEP 4

Assume Year 1 as base year:

① Set Real $GDP_1^F = Y_1$

$= \$200$

② Real $GDP_2^F = g^F \times Y_1$

$= 1.49 \times 200$

$= \$298.$